HOCHSCHULE NORDHAUSEN
University of Applied Sciences

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Term Paper

Lecture Scientific Practice

General conditions

- Look for the relevant scientific publications,
- evaluate them and
- summarise the findings on about 5 pages.

→ If you still have any doubts, please contact your supervisor

The editing period begins on 17.11.2022 and ends on 17.01.2023.
Send your term paper as a **pdf-document** via e-mail to your supervisor.

How to get started

1. Try to understand the given paper
2. 1st contact your supervisor and discuss your understanding of the paper
3. Start the research for related/controverse publications
4. 2nd contact to your supervisor to discuss the found literature/sources
5. Start writing

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Choosen Topics CES

01 - Determination of the range of 868MHz LoRa radio modules (Prof. Neitzke)

02 - Communication in the extremely low frequency band (ELF) 3Hz – 30Hz and super low frequency band (SLF) 30Hz – 300Hz (Prof. Neitzke)

03 - Light communication between two modules via a passive structure, (meteorological) clouds or the moon (Prof. Neitzke)

07 - Factors influencing the failure rate of electrolytic capacitors (Prof. Viehmann)

08 - Measurement arrangements and measurement results for the Schumann resonance (Prof. Viehmann)

09 - Impact of Maker Spaces on Engineering Education (Dipl.-Ing. Mielke)

11 - Integrated data model and structure for the asset administration shell in Industrie 4.0 (Dipl.-Ing. Mielke)

12 - Empowering Sustainable Consumption by Giving Back to Consumers the ‘Right to Repair’ (Dipl.-Ing. Brichet)

15 - Transient fault handling techniques for embedded processors (Prof. Schölzel)

16 - The greenest programming language (Dipl.-Ing. Tabatt)

17 - The optimal database for small computers (Dipl.-Ing. Tabatt)

19 - Scalable Distributed Data Anonymization for Large Datasets (Prof. Dotsenko)

20 - Analysis of Deep Learning Models for Anomaly Detection in Time Series IoT Sensor Data (Prof. Dotsenko)

Choosen Topics ERT

01 - Metal recycling (Prof. Rutz)

02 - Photovoltaic recycling (Prof. Rutz)

04 - Plastics recycling (Prof. Rutz)

05 - Rare earth materials (Prof. Rutz)

06 - Sorting and Recycling of Lightweight Packaging in Germany (Prof. Rutz)

07 - Hydrogen as an alternative energy carrier from aluminium waste (Prof. Rutz)

08 - The limits of recycling for its contribution to climate change mitigation (Prof. Rutz)

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Choosen Topics RES-1

06 - Bioenergy in the future (M.Eng. Vincent)

07 - Energy modelling and dynamic life cycle assessment (M.Eng. Vincent)

08 - Biomethane as alternative fuel (M.Eng. Vincent)

09 - Solar Collector Modelling (Dr.-Ing. Leibbrandt)

10 - Drainback Solar Collector System (Dr.-Ing. Leibbrandt)

11 - Energy systems modelling (M.Eng. Dhungel)

12 - High Efficiency Solar Flat Plate Collector Design (Dr.-Ing. Leibbrandt)

13 - Heat Transfer Calculation Approach (Dr.-Ing. Leibbrandt)

14 - Insulated Glass Solar Thermal Collector Design (Dr.-Ing. Leibbrandt)

15 - Hydrogen (M.Eng. Dhungel)

16 - Printed Solar Cells (M.Eng. Dost)

17 - Flexible Lithium Batteries (M.Eng. Dost)

18 - Multifunctional Batteries (M.Eng. Dost)

19 - Biogas (M.Eng. Dost)

20 - Heat Pumps (M.Eng. Dost)

21 - Data science in Energy analysis (M.Eng. Dhungel)

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Choosen Topics RES-2

22 - Modelling Tools (Dr.-Ing. Lustermann)

23 - Data Analysis (Dr.-Ing. Lustermann)

24 - Electric vehicle will overload the power grid! Myth? (M.Eng. Krishnan)

25 - Impact of E-mobility on household power consumption (M.Eng. Krishnan)

27 - Summary of available open-source tools for mobility load curve simulation (M.Eng. Krishnan)

28 - Alternative Fuels (Prof. Fischer)

29 - Fuel Cells (Prof. Fischer)

30 - Bioenergy: Fast pyrolysis (Prof. Fischer)

31 - Synthetic natural gas (Prof. Fischer)

35 - Simulation of a Scroll Expander (M.Eng. Senge)

39 - Drones in warfare (B.Eng. Reinhardt)

42 - Zero Energy Home (B.Eng. Reinhardt)

43 - Electric energy power supply of a typical IoT device by a solar cell and a storage battery (Prof. Neitzke)

44 - Electric energy power supply of a typical IoT device by a wind turbine and a storage battery (Prof. Neitzke)

Choosen Topics RES-3

45 - Energy pay back time of solar cells (Prof. Wesselak)

46 - What are the limits of pv-module efficiencies? (Prof. Wesselak)

47 - life-cycle assessment (LCA) of the GHG emissions of passenger cars (Prof. Wesselak)

48 - Methodology Development for Batteries Performances Analysis (M.Eng. Oberdorfer)

50 - Photovoltaic Tracking Systems (Prof. Wesselak)

51 - Potentials of hydrogen technologies in aviation (M.Eng. Rothe)

52 - The road to green air transport: climate change, technologies and user behaviour (M.Eng. Rothe)

53 - Possibilities of hydrogen storage - LOHC technology (M.Eng. Rothe)

54 - Challenges of icing on wind turbine rotor blades (M.Eng. Rothe)

56 - Vehicle to grid systems (M.Eng. Oberdorfer)

57 - Sustainable Mobility-Concepts: Bike-Sharing (B.Eng. Knahl)

58 - Recycling Process: Recycling Wind Turbine Blades (B.Eng. Knahl)

59 - Sustainable Mobility-Concepts: e-moped-Sharing (B.Eng. Knahl)

60 - Sustainable Mobility-Concepts: Sustainable Mobility Efficiency Index (B.Eng. Knahl)

63 - A review of energy storage types, applications and recent developments (M.Eng. Rathje)